Indian Institute of Technology Jodhpur

M.Tech. Composite Materials Technology Curriculum

Department of Materials Engineering

Course Structure: Semester wise distribution of credits for M.Tech. Composite Materials Technology students for the batches from Academic Year 2026-27.

I Semester						**II Semester**				
Cat	*Course Code*	*Course Title*	*L-T-P*	*Credits*		*Cat*	*Course Code*	*Course Title*	*L-T-P*	*Credits*
C	MTL7XXX	Introduction to Composite Materials	3-0-0	3		C	MTL7XXX	Composite Mechanics and Performance Evaluation	3-0-0	3
C	MTL7XXX	Composite Processing and Fabrication Technology	3-0-0	3		E	MTL7XXX	Elective	3-0-0	3
C	MTL7XXX	Characterization of Composite Materials	3-0-0	3		E	MTL7XXX	Elective	3-0-0	3
C	MTPXXX	Composite Materials Processing & Characterization Lab	0-0-2	1		E	MTL7XXX	Elective	3-0-0	3
E	MTL7XXX	Elective								
NG1	LAN7XX0	Technical Communication	1-0-0	S/X		NG2	OAL7XX0	Innovation and IP Management	1-0-0	S/X
			Total						*Total*	
T	MES7xxx	Summer Thesis								8
III Semester						**IV Semester**				
T	MTT7XX0	Thesis		16		T	MTT7XX0	Thesis		16
Toal				16			Total			16

S.No.	*Category*	*Course Category Title*	*Total Courses*	*Total Credits*
1	C	*COMPULSORY*	5	13
2	E	*ELECTIVES*	4	12
3	NG	*Non-Graded*	2	0
3	T	Thesis	**32+8**	32+8
		Total		**65**

***for Direct students and students with different background.**

Program compulsory (C)	**Electives (E)**
Introduction to Composite Materials, Composite Processing and Fabrication Technology, Characterization of Composite Materials Composite Materials Processing & Characterization Lab, Composite Mechanics and Performance Evaluation	Any from the Elective list

Department of Materials Engineering

Program Name M.Tech. Composite Materials Technology

1. The following four courses are compulsory for M.Tech. students of the department:

    1. **Introduction to Composite Materials**
    2. **Composite Processing and Fabrication Technology**
    3. **Characterization of Composite Materials**
    4. **Composite Materials Processing & Characterization Lab**
    5. **Composite Mechanics and Performance Evaluation**

2. M.Tech. students shall have the option to choose four courses from the following list of courses:

    1. **Polymer Matrix Composites**
    2. **Ceramic Matrix Composites**
    3. **Sustainable Composites**
    4. **AI/ML and Big data for composites**
    5. **Lightweight Composites for Automotive and Aerospace Applications**
    6. **Computational Modelling of Composites**
    7. **Biomedical and Healthcare Composites**
    8. **Composites for Energy Harvesting and Storage**
    9. **Machining and Tribology of Composites**
    10. **Advanced Composite Design & Analysis**
    11. **Composites for Defence Technology**
    12. **Functional Nanocomposites and Hybrid Composites**

The syllabus and checklists for the new proposed courses are given below

Course Title	Introduction of Composite Materials	Course No.	MTLXXX
Department	Materials Engineering	Structure (L-T-P-D)	3-0-0
Offered for	B.Tech. (Final Year)/M.Tech./Ph.D. (Course Work)	Type	Compulsory
Prerequisite	-		

Objectives

The objectives of this course are to:

1. Build fundamental understanding of composite materials (types, constituents, interfaces) and their processing–structure–property relationships for engineering applications.
2. Develop a comprehensive foundation in composite material systems, emphasizing constituent interactions and processing–microstructure–property correlations.

Learning Outcomes

After completing this course, students will be able to:

1. Enable students to analyse, predict properties, and select appropriate composites (PMCs, MMCs, CMCs) based on materials, processing techniques, and performance requirements.
2. Equip students to quantitatively predict behaviour (e.g., rule of mixtures) and make informed material selection decisions for real-world engineering applications.

Contents:

Fundamental of Composites (11 lectures) Concepts, Classification of composites, Type of matrix and reinforcement, Basic composites theory, Volume fraction, Principles of heterogeneous composites, Role of wettability, Interfacial bonding mechanisms, Surface Energy, Sizing chemistry, basic of mechanical properties, Types of Fiber Architectures and Their practical implications, Standards

Polymer Matrix Composites (PMCs) (10 lectures) Classification, Thermoplastic composites, thermoset composites, Rubber matrix composites, Plasticization, Glass transition temperature, Fibre PMCs, Reinforcing fillers and additives, Percolation threshold, processing techniques, Critical fibre length, Advanced Matrix Systems and Cure Chemistry, Crystallization, Cure monitoring, Ply orientation effects on mechanical properties.

Ceramics Matrix Composites (CMCs) (10 lectures) Introduction to CMCs, Engineering ceramics, Types of reinforcement, Processing of ceramics, Properties & applications, Toughening mechanisms, Concepts of statics/reliability & probability of failure (Weibull modulus), Alumina matrix composites, SiC whisker reinforced, Zirconia toughened alumina, Glass-ceramic matrix composites, Carbon-carbon composites.

Metal Matrix Composites (MMCs) (8 lectures) Fundamentals, Types of metal matrices and reinforcements, Processing of MMC, In-situ composites, Particle bonding, Distribution in MMCs, Strengthening mechanisms, Microstructure, Defect Formation, Applications

Textbook

1. Chawla K.K., "Composite Materials: Science and Engineering", 3rd Ed., Springer, 2012. [Particulate/MMC/CMC; microstructure–property relationships]
2. Hull D. & Clyne T.W., "An Introduction to Composite Materials", 2nd Ed., Cambridge University Press, 1996. [Concise; physical insight into fibre–matrix mechanics]
3. ASTM D3039 — Standard Test Method for Tensile Properties of Polymer Matrix Composite Materials.

Reference

1. Barbero E.J., "Introduction to Composite Materials Design", 3rd Ed., CRC Press, 2018. [Units 1–3 — design-oriented; CADEC software; applications]

Course Title	Composite Processing and Fabrication Technology	Course No.	MT7XXX
Department	Materials Engineering	Structure (L-T-P-D)	3-0-0 [3]
Offered for	B.Tech. (Final Year)/M.Tech./Ph.D. (Course Work)	Type	Compulsory
Prerequisite	-		

Learning Objectives:

The objectives of this course are to:

1. Develop fundamental understanding of composite materials manufacturing methods, selection criteria, and processing techniques including open-mould, closed-mould, thermoset, thermoplastic, and advanced manufacturing.
2. Provide hand-on experience of composite fabrication techniques and processing methods through laboratory experiments.

Learning Outcomes:

After completing this course, students will be able to:

1. Select and apply appropriate composite manufacturing processes based on material system, geometry, production requirements, and cost considerations.
2. Perform composite fabrication using different techniques, interpret experimental results, and optimise process parameters to achieve the desired properties.

Course Contents:

Fundamentals of composites manufacturing, open-mould composite processing (6 lectures): Fundamentals of material selection process, design for manufacturing, hand lay-up process and procedures, spray-up process and methods, vacuum bagging, vacuum-assisted methods, defects, and tools.

Prepreg and thermoplastic composite processing (8 lectures): Prepreg materials, autoclave processing, compression moulding, film techniques, extrusion, two-roll mill, recycling, reprocessing and thermoforming.

Closed mould and liquid composite moulding processes (6 lectures): Resin transfer moulding, Vacuum assisted resin transfer moulding, Resin Transfer Molding, Reaction injection moulding, Resin flow behaviour, Void formation, and Process parameters.

Metal matrix composites processing (5 lectures): Metal matrix composites, manufacturing processes, dispersion-hardened and particle composites, layer composites and infiltration methods.

Ceramic matrix composites processing (8 lectures): Ceramic Matrix composites and processing, Liquid phase processing, chemical processing routes, solid-state processing, conventional sintering, advanced pressure-assisted sintering methods, and densification mechanism.

Continuous and advanced composite manufacturing (6 lectures): Pultrusion process, filament winding, winding patterns, fiber placement (AFP) and automated tape laying (ATL), defects, additive manufacturing of composites, and recent developments in composites manufacturing.

Textbook

1. Manufacturing processes for advanced composites by FC Campbell Jr (2003).
2. Composites manufacturing: materials, product, and process engineering by S. Mazumdar, (2001).
3. Composite materials and processing by M. Balasubramanian, (2013).

References:

1. Kamal M.R. & Sourour S., Kinetics and Thermal Characterisation of Thermoset Cure, Polymer Engineering & Science, 1973. [Kamal-Sourour cure kinetics model]
2. Gutowski T.G. (Ed.), Advanced Composites Manufacturing, Wiley, 1997. [Process modelling; void evolution; compaction mechanics]
3. ASTM D5687: Preparation of Flat Composite Panels with Processing Guidelines for Specifying Parameters.

Online:

1. NPTEL Lecture on Manufacturing of Polymer Matrix Composites, IIT Kanpur: https://nptel.ac.in

Course Title	Characterization of Composite Materials	Course No.	MTL7XXX
Department	Materials Engineering	Structure (L-T-P-D)	3-0-0
Offered for	M.Tech./Ph.D. (Course work)	Type	Compulsory
Prerequisite	-		

Learning objectives:

The objectives of this course are to:

1. Introduce the fundamental concepts and techniques used for the characterization of composite materials.
2. Develop a comprehensive understanding of physical, thermal, mechanical, and microstructural characterization techniques for composite materials.
3. Provide hands-on training in experimental methods, data acquisition, and analysis for composite materials characterization.

Learning outcomes:

After completing this course, students will be able to:

1. Select and apply appropriate characterization techniques across physical, thermal, mechanical, and microstructural domains of composite materials.
2. Critically analyze and interpret experimental data to correlate composite structure, processing conditions, and overall performance.

Course Content:

Physical characterization (6 lectures): Introduction to physical characterization, measurement of density, fiber volume fraction, void volume content, porosity, shape retention and dimensional stability, and effect of manufacturing parameters.

Thermal characterization (11 lectures): Introduction to thermal behaviour, glass transition temperature, differential scanning calorimetry (DSC), thermogravimetric Analysis (TGA), decomposition behaviour, estimation of filler or fiber content, dynamic mechanical analysis (DMA), viscoelastic behaviour, thermomechanical analysis (TMA), coefficient of thermal expansion (CTE), and measurement of thermal conductivity and thermal diffusivity.

Mechanical characterization (11 lectures) : Mechanical behaviour of composite materials, tensile testing, flexural testing, compression testing, interlaminar shear strength, impact testing methods, fatigue testing methods, fracture toughness of composites, failure mechanisms, indentation testing, and failure data analysis.

Microstructural characterization & Phase Characterization (11 lectures): Microstructural analysis in composite materials, optical microscopy, scanning electron microscopy (SEM), transmission electron microscopy (TEM), atomic force microscopy (AFM), X-ray Diffraction (XRD), fourier transform infrared spectroscopy (FTIR), Raman Spectroscopy, and surface roughness measurement techniques.

Textbook:

1. Experimental Characterization of Advanced Composite Materials by Leif A. Carlsson, Donald F. Adams, and R. Byron Pipes, (1st edition), CRC Press, 1994.
2. Characterization of Composite Materials by Hatsuo Ishida, (1st edition), Butterworth-Heinemann, 1984.
3. Failure Analysis and Fractography of Polymer Composites by Emile Greenhalgh, Woodhead Publishing, 2009.

References:

1. Ishida, Hatsuo, ed. Characterization of composite materials. Elsevier, 2013.
2. Sachdeva, Amit, Pramod Kumar Singh, and Hee Woo Rhee, eds. Composite materials: properties, characterisation, and applications. CRC Press, 2021.

Online:

1. NPTEL Lecture: Characterization of Polymers, Elastomers and Composites by Prof. Santanu Chattopadhyay, IIT Kharagpur. https://nptel.ac.in/courses/103105219

Course Title	Composite Materials Processing & Characterization Lab	Course No.	MT7XXX
Department	Materials Engineering	Structure (L-T-P-D)	0-0-2
Offered for	M.Tech.	Type	Compulsory
Prerequisite	-		

Learning Objectives:

The objectives of this lab are to:

1. Develop fundamental understanding of composite material processing, fabrication methods, and advanced manufacturing techniques for polymer, ceramic, and metal matrix composites.
2. Provide hands-on experience in composite fabrication, specimen preparation, and characterization techniques for evaluating mechanical, thermal, physical, and microstructural properties.

Learning Outcomes:

After completing this course, students will be able to:

1. Select and perform suitable composite manufacturing and specimen preparation techniques based on material systems, processing requirements, and application needs.
2. Conduct and interpret mechanical, thermal, physical, and microstructural characterization of composites, and correlate processing parameters with material performance and failure behavior.

Course Content

Manufacturing of composites using Compression molding, Autoclave, VARTM, Effect of moulding temperature and pressure on thermoplastic composite properties, Electrospinning, Ball milling for ccompaction and sintering, Filament Winding Processing, Density measurement in composite materials, Fiber volume fraction and void content calculation in FRP composites, DSC for glass transition and thermal behaviour, TGA for decomposition and filler/fiber content estimation. DMA/TMA for viscoelastic properties and CTE, Tensile testing, Flexural testing, ILSS testing, Impact testing (Charpy/Izod) and damage analysis, Microscopy for microstructural analysis of composites, FTIR and XRD,

Textbook

1. Manufacturing processes for advanced composites by FC Campbell Jr (2003).
2. Composites manufacturing: materials, product, and process engineering by S. Mazumdar, (2001).
3. Composite materials and processing by M. Balasubramanian, (2013).
4. Experimental Characterization of Advanced Composite Materials by Leif A. Carlsson, Donald F. Adams, and R. Byron Pipes, (1st edition), CRC Press, 1994.
5. Characterization of Composite Materials by Hatsuo Ishida, (1st edition), Butterworth-Heinemann, 1984.
6. Failure Analysis and Fractography of Polymer Composites by Emile Greenhalgh, Woodhead Publishing, 2009.

References:

1. Kamal M.R. & Sourour S., Kinetics and Thermal Characterisation of Thermoset Cure, Polymer Engineering & Science, 1973. [Kamal-Sourour cure kinetics model]
2. Gutowski T.G. (Ed.), Advanced Composites Manufacturing, Wiley, 1997. [Process modelling; void evolution; compaction mechanics]

Course Title	Composite Mechanics and Performance Evaluation	Course No.	MTL7XXX
Department	Materials Engineering	Structure (L-T-P-D)	3-0-0
Offered for	M.Tech./Ph.D. (Course Work)	Type	Compulsory
Prerequisite	-		

Learning Objectives:

The objectives of this course are to:

1. Develop a fundamental understanding of mechanics of FRP composites, including stress–strain behavior, failure theories, and basic mechanics concepts.
2. Analyze mechanical behavior of composite lamina and laminates using both micromechanical and macromechanical approaches, including classical lamination theory.
3. Evaluate failure mechanisms and design composite laminates by applying appropriate failure criteria, damage analysis, and standard testing methods for engineering applications.

Learning outcomes:

After completing this course, students will be able to:

1. Apply mechanics concept to composite materials to evaluate stress–strain analysis, deformation behavior, and failure prediction under different loading conditions.
2. Perform micro and macro mechanical analysis of composite lamina and laminates to evaluate elastic properties and stress–strain relationships.
3. Select and assess composite laminates for engineering applications, including failure analysis, damage progression evaluation, and selection of suitable materials and performance requirements**.**

Course Content:

Introduction to solid mechanics and FRP composites (10 lectures): Fundamental stress & strain, analysis of stress and strain, Hooke's law for different types of material (anisotropic material, monoclinic material, orthotropic material, transversely isotropic material, isotropic material), axial loading and torsion, beams and bending, strain energy and failure theories, theoretical fracture strength, basic terminologies related to FRP composites.

Macromechanical analysis of a lamina (7 lectures): Hooke's law for a two-dimensional unidirectional lamina (plane stress assumption, reduction of Hooke's law in three dimensions to two dimensions, relationship of compliance and stiffness matrix to engineering elastic constants of a lamina), Hooke's law for a two-dimensional angle lamina, strength failure theories of an angle lamina.

Micro-mechanical analysis of a lamina (7 lectures): Introduction, evaluation of the four elastic constants using mechanics of materials approach, exact solution, the halpin-tsai equations, ultimate strengths of a unidirectional lamina, coefficients of thermal expansion, coefficients of moisture expansion.

Macromechanical analysis of laminates (8 lectures): Common laminate definitions, classical lamination theory, stress-strain relations for a laminate, shear-extension coupling, bending-extension coupling, bend-twist coupling, special cases of laminates (isotropic laminate, specially orthotropic laminate, anisotropic laminate, symmetric laminates, antisymmetric laminates, antisymmetric cross-ply laminates), laminate strength analysis procedure, interlaminar stresses.

Failure analysis, and design of laminates (7 lectures): Failure theories of lamina and laminates, fracture modes in composites, design of a laminated composite, design examples, mechanical design issues, standard mechanical testing methods of composites, damage in composites, damage progression mechanics, composite structures and their applications.

Textbook

1. Mechanics of Composite Materials, 2nd Edition, CRC Press by Robert M Jones (1999).
2. Mechanics of Composite Materials, 2nd Edition, Taylor and Francis by Autar Kaw (2005).
3. Engineering Mechanics of Composite Materials, 2nd Edition Oxford University Press by I M Daniel and O Ishai (2006).

Online:

1. NPTEL: Mechanics Of Fiber Reinforced Polymer Composite Structures by Prof. Debabrata Chakraborty, IIT Guwahati. https://nptel.ac.in/courses/112103308
2. NPTEL: Mechanics of Composite Materials and Structures By Prof. Mohammed Rabius Sunny, IIT Kharagpur. https://nptel.ac.in/courses/101105662

Course Title	Polymer Matrix Composites	Course No.	MTL7XXX
Department	Materials Engineering	Structure (L-T-P-D)	3-0-0
Offered for	B.Tech. (Final Year)/M.Tech./Ph.D.	Type	Elective
Prerequisite	-		

Learning objectives:

1. Develop a fundamental understanding of the constituents, interfaces, processing routes, and mechanical behaviour of polymer matrix composites.
2. Introduce students to industrial composite manufacturing techniques and the relationship between process parameters and final component quality.

Learning outcomes:

Students are expected to:

1. Classify composite materials and identify the role of matrix, reinforcement and interface in governing composite performance.
2. Explain the chemistry and processing characteristics of major thermosetting and thermoplastic matrices used in PMCs.

Contents:

Introduction to Polymer Matrix Composites [4 lectures]: Definition and classification of composites, constituents: matrix, reinforcement, interface, applications, historical development and market overview.

Reinforcements [5 lectures]: Glass fibers, carbon/graphite fibers, aramid, natural fibers, Kevlar, basalt, woven, non-woven and braided architectures, particulate and short-fiber reinforcements, sizing and coupling agents.

Matrix Materials [11 lectures]: Thermosetting matrices, Thermoplastic matrices, toughening strategies, selection criteria for matrix–reinforcement combinations, Polymer Nanocomposites

Interface and Interphase [4 lectures]: Wetting and adhesion, micromechanics of stress transfer, surface treatments for reinforcing filler, sizing chemistry.

Polymers Processing and technology [8 lectures]: Additives and Filler, Lubricants, UV-Stabilizer, Plasticizer, Chain Extenders, Processing aids, Compatibilizers, Extrusion, Sheet casting, lay-ups

Testing and standards [7 lectures]: Rule of mixtures, thermal testing, mechanical testing, laminate theory, failure modes, fatigue and fracture toughness of PMCs, testing standards.

Textbook

1. Hull, D. and Clyne, T.W., (1996), An Introduction to Composite Materials, 2nd Edition, Cambridge University Press.
2. Mallick, P.K., (2007), Fiber-Reinforced Composites: Materials, Manufacturing and Design, 3rd Edition, CRC Press.

References:

1. Chawla, K.K., (2012), Composite Materials: Science and Engineering, 3rd Edition, Springer.
2. Barbero, E.J., (2018), Introduction to Composite Materials Design, 3rd Edition, CRC Press.
3. Agarwal, B.D., Broutman, L.J. and Chandrashekhara, K., (2017), Analysis and Performance of Fiber Composites, 4th Edition, Wiley.
4. Advani, S.G. and Hsiao, K.-T. (Eds.), (2012), Manufacturing Techniques for Polymer Matrix Composites, Woodhead Publishing.

Online:

1. NPTEL: Composite Materials, IIT Madras, Prof. M.S. Sivakumar, https://nptel.ac.in/courses/112106153.
2. MIT OCW: Introduction to Composite Materials, 3.35, https://ocw.mit.edu.
3. Coursera: Composite Materials, Duke University, https://www.coursera.org.

Course Title	Ceramic Matrix Composite	Course No.	MTL7XXX
Department	Materials Engineering	Structure (L-T-P-D)	3-0-0
Offered for	B.Tech. (Final Year)/M.Tech./Ph.D.	Type	Elective
Prerequisite	-		

Learning Objectives:

The Instructor will provide:

1. A basic understanding of ceramic matrix composites (CMCs), their processing and characterization techniques.
2. Background to different processing techniques for CMC fabrication, interpret mechanical, thermal, and environmental behavior and apply knowledge to real-world engineering applications

Learning Outcomes:

The students are expected to have the ability to:

1. Understand the structure, constituents, and processing methods of Ceramic Matrix Composites (CMCs), and evaluate how processing influences their properties and performance.
2. Analyze failure mechanisms and behavior in extreme environments and apply this knowledge to the basic design of CMC components for engineering applications.

Contents

Introduction (5 lectures) : Introduction to Ceramics, Definition and classification of composites, Introduction to Ceramic matrix composites (CMCs), Advantages of CMCs over monolithic ceramics, constituents of CMCs: Ceramic matrices and reinforcements: (continuous/discontinuous fibers, whiskers), Interface and interphase engineering.

Fabrication and Processing of CMCs (10 lectures): Slurry infiltration/High pressure sintering technique (SI-HPS), High temperature (HT) processing, Chemical vapor infiltration (CVI), Polymer impregnation and pyrolysis (PIP process), Reactive melt infiltration (RMI - process), Processing defects and quality control, thermal spray processing for deposition of CMC coatings.

Microstructure and Mechanical Properties (10 lectures): Microstructure (fibre architecture, porosity), Mechanical properties, Role of Interface in load transfer in CMCs, Toughening and strengthening mechanisms in CMCs, Crack deviation mechanism, Fracture mechanics of CMCs, Tailoring fiber/matrix interfaces, Self- crack-healing behavior in ceramic matrix composites, Weak interface composites (WIC), Weak matrix composites (WMC), Statistics and probability of failure in CMCs (Weibull Modulus).

Thermal Stresses in CMCs (4 lectures): CTE mismatch in composites, Thermal Residual Stresses in CMCs (origin, analytical models), Thermal Stress-Induced Damage (matrix cracking, Fiber/matrix debonding), Thermal shock resistance, Thermo-mechanical fatigue (TMF).

Testing methods for CMCs (5 lectures): Testing techniques for CMC materials; CMC testing standards, ASTM C28 committee standards, Non-destructive testing techniques, Ultrasonic analysis, Thermography, X- Ray Analysis.

Applications of CMCs (5 lectures): CMCs for structural applications, CMCs for friction applications, CMC for metal cutting applications, glass–ceramic composites for microelectronics, CMC for space and aeronautical applications; CMC for nuclear applications.

Textbook:

1) K. K. Chawala, (2003), Ceramic Matrix Composites, 2nd Edition, Springer US.

2) N.P Bansal and J. Lamon, (2015), Ceramic Matrix Composites: Materials, Modeling and Technology, John Wiley & Sons.

Reference Book:

1) Walter Krenkel, (2008), Ceramic Matrix Composites- Fiber Reinforced Ceramics and their Applications, Wiley-VCH.

2) I.M. Low, (2014), Advances in ceramic matrix composites, 2nd Edition, Woodhead Publishing Ltd.

Course Title	Sustainable Composites	Course No.	MTL7XXX
Department	Materials Engineering	Structure (L-T-P-D)	3-0-0
Offered for	B.Tech. (Final Year)/M.Tech./Ph.D.	Type	Elective
Prerequisite	-		

Learning Objectives:

Upon successful completion of this course, students will be able to:

1. Characterise bio-based reinforcements (jute, flax, hemp, sisal, coir, bamboo) and bio-derived matrix systems Quantify the environmental performance of composite systems using life cycle assessment and compare bio-composite alternatives to conventional glass/carbon PMCs.
2. Evaluate the techno-economic and regulatory context of sustainable composites in construction, packaging, consumer goods, and automotive sectors.

Learning Outcomes:

Students are expected to:

1. Measure and compare specific mechanical properties of UD jute, flax, and E-glass composites and interpret variability using Weibull statistics.
2. Develop a green product design brief specifying bio-composite material stack, process, recyclability pathway, and compliance with EN 13432 or ISO 14001.

Contents:

Introduction to sustainability in materials engineering [6 Lectures]: Triple bottom line (environmental, economic, social), Material flows and resource scarcity, Energy and carbon footprint of composites, Introduction to Circular Economy, Sustainability metrics and indicators, Global frameworks and policy drivers.

Life Cycle Assessment and Tools [7 Lectures]: Overview of Life Cycle Assessment (LCA), Goal and scope definition, Life Cycle Inventory (LCI), Life Cycle Impact Assessment (LCIA), Interpretation of LCA results, LCA software tools, Case study: LCA of glass vs natural fiber composites.

Bio-based and Natural Fiber Composites [7 Lectures]: Introduction to bio-based composites, Natural fibers, Biopolymers, Fiber-matrix compatibility, Processing Techniques, Performance evaluation, Cellulose, Applications and limitations.

Recyclable and Circular Composites [7 Lectures]: Challenges in recycling conventional composites, Thermoplastic composites, Recyclable thermosets, Mechanical recycling methods, Chemical recycling, Design for recycling and disassembly, Industrial case studies.

Emerging Sustainable Composites [7 Lectures]: Introduction to nanocomposites, Carbon-based nanomaterials, Multi-scale reinforcement strategies, Self-healing composites, Smart and multifunctional composites, AI/ML in materials design.

Applications and Case Studies [5 Lectures]: Automotive sector, Aerospace sector, Renewable energy systems, Water and environmental applications, Infrastructure and future directions

Textbook

1. Faruk O. et al., "Biocomposites Reinforced with Natural Fibers: 2000–2010", Progress in Polymer Science, 37, 2012. [Units 1–2 — definitive review of natural fibre composite properties and processing]
2. Baillie C. (Ed.), "Green Composites: Polymer Composites and the Environment", Woodhead, 2004. [Unit 3 — LCA, recycling, sustainability case studies]
3. John M.J. & Thomas S. (Eds.), "Natural Fibres, Biopolymers and Biocomposites", CRC Press, 2008. [Units 1–2 — fibre types, surface modification, bio-matrices]
4. Corbière-Nicollier T. et al., "Life Cycle Assessment of Biofibres Replacing Glass Fibres", Resources, Conservation & Recycling, 2001. [Unit 3 — LCA comparison benchmark study]

References:

1. Mohanty A.K., Misra M. & Drzal L.T. (Eds.), "Natural Fibers, Biopolymers and Biocomposites", CRC Press, 2005.
2. Pickering K.L. (Ed.), "Properties and Performance of Natural-Fibre Composites", Woodhead, 2008.
3. OpenLCA Software: https://www.openlca.org — Free LCA modelling tool
4. NPTEL — Green and Eco-Friendly Composites: https://nptel.ac.in

Course Title	AI/ML and Big Data for Composites	Course No.	MTL7XXX
Department	Materials Engineering	Structure (L-T-P-D)	3-0-0
Offered for	B.Tech. (Final Year)/M.Tech./Ph.D.	Type	Elective
Prerequisite	-		

Objectives:

1. To orient students towards usage of data driven approaches and machine learning algorithms for solving problems in the field composite materials.
2. To train students with hands-on learning experiences on various case studies involving prediction, classification and generative modeling of composite materials design and properties.

Learning outcomes:

1. Ability to apply modern AI/ML tools in analyzing and solving problems related to composite materials.
2. Ability in handling, analysis, interpretation and generation of big data in composite materials domain.

Course Content

Module 1: Big Data in Composite Materials (6 Lectures)

Dataset creation from experiments, simulations, and literature data mining; statistical analysis; correlations, handling missing data, dataset imbalances, dimensionality reduction and visualization of high-dimensional datasets, feature generation and elimination techniques, data leakage in machine learning workflows, data distributions during training and inference, and efficient data loading techniques for large datasets. Hands on training session on programming in python, data types, functions, loops, and importing python libraries, Data handling, visualization, heatmaps, correlations in datasets.

Module 2: Machine Learning Techniques (13 lectures)

Supervised, unsupervised, reinforcement, and generative learning paradigms, bias and variance trade-off, underfitting and overfitting, classification, regression and clustering problems, linear and logistic regression, ridge and lasso regression, code demonstration on application of supervised learning in damage mode identification of composites. Implementing linear models for composite materials through scikit learn, effect of regularization.

Tree-based and kernel methods including random forests and support vector machines; ensemble learning approaches such as bagging and boosting; model interpretation techniques including feature importance analysis, code demonstration on application of regression models in physical property prediction of polymer composites. Bayesian optimization, surrogate modeling and inverse materials design.

Module 3: Neural Networks and Deep Learning (13 lectures)

Architecture of artificial neural networks, activation functions, training using backpropagation, and optimization algorithms. Model explanation by SHAP and LIME, Fatigue life prediction of polymer composites using ANN

Convolutional neural networks (CNNs) including convolution operations and pooling; advanced architectures such as U-Net and ResNet, application of CNN as surrogate model for finite element model for the prediction of effective stiffness and Poisson' ratio, demonstration of defects detection using U-Net/ResNet.

Recurrent neural networks (RNNs) and their variants including Long Short-Term Memory (LSTM) and Gated Recurrent Units (GRU), prediction of spatiotemporal evolution of thermochemical ablation of composites

Module 4: Generative and Transformer-Based Models (7 lectures)

Generative modeling approaches including Generative Adversarial Networks (GANs) and Variational Autoencoders (VAEs); modern architectures such as pretrained transformers and vision transformers, application of generative models in inverse design of composite materials

Textbooks

1. Géron, A., *Hands-on Machine Learning with Scikit-Learn, Keras & TensorFlow*, O'Reilly Media, 2nd edition, 2019.
2. Faroughi, S. A., Fernandes, C. B. P., Scientific Machine Learning for Polymeric Materials, Polymers, 2025, ISBN 978-3-7258-5840-8

Reference

1. Sanjay, M. R., Madhushri, P., Siengchin, S., Kushvaha, V., Machine Learning Applied to Composite Materials, Springer Nature Singapore, 2022.
2. NPTEL videos on *Artificial Intelligence and Machine Learning in Materials Engineering* by Prof. Krishanu Biswas, IIT Kanpur. [link]

Course Title	Lightweight Composites for Automotive and Aerospace Applications	Course No.	MTL7XXX
Department	Materials Engineering	Structure (L-T-P-D)	3-0-0
Offered for	B.Tech (Final Year)/M.Tech./Ph.D.	Type	Elective
Prerequisite	-		

Learning Objectives:

The objectives of this course are to:

1. Develop a fundamental understanding of lightweighting strategies using composites and their role in improving performance and efficiency in automotive and aerospace components.
2. Analyse and design composite materials by considering various engineering constraints in automotive and aerospace applications.
3. Expose students to available manufacturing processes, material integration, and certification requirements for composite structures used in industry.

Learning outcomes:

After completing this course, students will be able to:

1. Apply material selection strategies to choose suitable composites for automotive and aerospace applications.
2. Perform structural design and analysis of composite components by evaluating stiffness, strength, impact response, and damage tolerance under different loading conditions.
3. Select appropriate manufacturing processes, implement quality control techniques, and identify certification requirements for composite structures.

Course Content:

Lightweighting Strategies and Material Systems in Automotives (5 lectures): Introduction, lightweighting strategies, mass reduction vs performance trade-offs, role of composites in IC Engine and electric vehicles (EVs), material substitution approaches, FRPs for automotive applications, Composite selection for automotive environments, comparison with conventional materials, limitations and industrial challenges.

Engineering Design of Automotive Structures (8 lectures): Design philosophy, stiffness-driven and strength-driven design, material selection using performance indices, stiffness optimization of beams and panels, design of body-in-white (BIW), crashworthiness and energy absorption mechanisms, damage and failures, design constraints, case studies on composite body panels and bumper beams.

Processing and Integration of Automotive Composites (6 lectures): Manufacturing requirements for high-volume production, thermoset vs thermoplastic composites for automotive use, cycle time and cost considerations, composite–metal integration in multi-material designs, defect characterization and quality control, non-destructive testing (NDT) methods, case studies on body panels and leaf springs.

Introduction to Aerospace Composite Materials (5 lectures): Material requirements, FRP composites in aircraft and space structures, high-performance laminated composites, sandwich structures, braided and 3D woven composites, auxetic composites, carbon–carbon composites, multiscale composites, mechanical performance, advantages and limitations of composites in aerospace applications.

Design and Performance of Aerospace Composites (8 lectures): Design requirements, laminate and sandwich structures, damage mechanisms, damage tolerance and fail-safe design, crash and impact resistance, thermal and environmental effects, structural health monitoring (SHM), self-sensing composites, self-healing composites and their mechanisms, joining techniques for composite structures, durability and life prediction, case studies on aircraft wings and fuselage components.
Processing, quality control and certification of aerospace composites (7 lectures): Processing requirements for aerospace-grade composites, manufacturing techniques, defects, inspection and quality control procedures, NDT techniques, repair and maintenance of composite structures, certification challenges and standardization issues, industrial applications and case studies in aerospace components.

Textbook

1. Automotive Composites: Materials, Design, and Manufacturing by P.K. Mallick, CRC Press, 2025.
2. Advanced composite materials for aerospace engineering: processing, properties and applications by Sohel Rana, and Raul Fangueiro, Woodhead Publishing, 2016.
3. Composite materials in aerospace design by Zagainov GI, Lozino-Lozinski GE, Springer Science & Business Media, 2012

References:

1. Mangalgiri, P. D, Composite materials for aerospace applications, Bulletin of Materials Science 22, no. 3 (1999): 657-664.
2. Gopal, K. V. N, Product design for advanced composite materials in aerospace engineering, In Advanced composite materials for aerospace engineering, pp. 413-428. Woodhead Publishing, 2016.

Course Title	Computational Modelling of Composites	Course No.	MTL7XXX
Department	Materials Engineering	Structure (L-T-P-D)	3-0-0
Offered for	M.Tech./Ph.D.	Type	Elective
Prerequisite	Composite Mechanics and Performance Evaluation		

Learning Objectives:
The instructor will:
1. provide an overview of the mechanics and constitutive behaviour of composite materials
2. develop finite element formulations and computational procedures for laminated composite structures
3. introduce computational approaches for nonlinear analysis, damage, and delamination modelling of composites

Learning Outcomes:
The students are expected to have the ability to:
1. formulate constitutive and laminate theories for composite materials and structures
2. perform finite element analysis of laminated composite plates and shells
3. introduced to computational modelling methods of damage and delamination in composites

Contents

Mechanics of Orthotropic Lamina [Lectures 8]: Brief review of solid mechanics, constitutive equations for orthotropic materials, transformation of stress and strain, transformed reduced stiffness matrix, thermoelastic behaviour, constitutive modelling of composite laminae

Mechanics of Laminated Composite Plates: [Lectures 8]: Kinematics of laminated plates, Classical Lamination Theory, laminate constitutive equations, A-B-D matrices, laminate response under mechanical, first-order shear deformation theory, higher-order laminate theories

Finite Element Formulation for Composite Structures [Lectures 11]: Variational principles, weak formulation, finite element discretization, one-dimensional and two-dimensional finite elements, isoparametric formulation, numerical integration, plate finite elements, finite element modelling of laminates, layerwise and equivalent single-layer approaches, nonlinear finite element formulation, Newton–Raphson method, convergence criteria, computational implementation using ABAQUS

Introduction to Fracture, and Delamination Modelling [Lectures 7]: Failure mechanisms in laminates, failure criteria, continuum damage mechanics, progressive damage modelling, cohesive zone modelling, computational fracture mechanics, delamination analysis

Introduction to Advanced Computational Methods [Lectures 5]: Computational and analytical homogenization, Introduction to multiscale computational modelling, Introduction to viscoelastic modelling of composites

Textbooks
1. Finite Element Analysis of Composite Materials — Ever J. Barbero, CRC Press, Taylor & Francis Group.
2. Mechanics of Composite Materials — Robert M. Jones , CRC Press, Taylor & Francis Group.
3. Mechanics of Laminated Composite Plates and Shells — J. N. Reddy , CRC Press, Taylor & Francis Group.

Reference Books
1. Nonlinear Finite Elements for Continua and Structures — Ted Belytschko, Wiley.

Online Course Material
1. Talukdar S., Plates and Shells, NPTEL Course Material, Department of Civil Engineering, Indian Institute of Technology Guwahati, https://archive.nptel.ac.in/courses/105/103/105103209/
2. Mechanics of Composite Materials, NPTEL Course Material, Department of Aerospace Engineering, Indian Institute of Technology Kanpur, https://nptel.ac.in/
3. Advanced Finite Element Analysis, NPTEL Course Material, Department of Civil Engineering, Indian Institute of Technology Roorkee, https://nptel.ac.in/
4. Composite Structures, OpenCourseWare, Massachusetts Institute of Technology, https://ocw.mit.edu/

Course Title	Biomedical and Healthcare Composites	Course No.	MTL7XXX

Department	Materials Engineering	Structure (L-T-P-D)	3-0-0
Offered for	B.Tech. (Final Year)/M.Tech./Ph.D.	Type	Elective
Prerequisite	-		

Learning Objectives:

The objectives of this course are to:

1. Introduce students to composite materials specifically engineered for biomedical and healthcare applications.
2. Develop understanding of biocompatibility requirements, regulatory frameworks, and design principles for implantable and non-implantable medical composites.

Learning Outcomes:

After completing this course, students will be able to:

1. Classify biomedical composites and evaluate their biocompatibility and mechanical suitability for specific clinical applications.
2. Design and select composite systems for orthopaedic, dental, cardiovascular, and tissue engineering uses, and critically assess regulatory and safety requirements.

Contents

Introduction to Biomedical Composites (6 lectures): Overview of materials in biomedical applications; Classification of biomedical composites, Biocompatibility criteria and ISO/ASTM standard, In vitro and in vivo evaluation strategies; Regulatory pathways for medical devices.

Orthopaedic and Bone Composites (7 lectures): Bone structure and mechanical properties, Hydroxyapatite-polymer composites, orthopaedic implants, Load bearing, Wear and corrosion behaviour.

Dental and Craniofacial Composites (6 lectures): Dental composite resins, shrinkage stress; Fibre-reinforced polymer composites for dental prostheses, Zirconia-based dental composites, Craniofacial reconstruction materials, Long-term clinical performance and wear mechanisms.

Soft Tissue and Cardiovascular Composites (6 lectures): Hydrogels, Composite vascular grafts and compliance matching, Wound dressing composites, Electrospun nanofibre composite membranes, Suture and fixation device composites.

Drug Delivery and Smart Biomedical Composites (7 lectures): Polymer-ceramic composite drug delivery matrices, Stimuli-responsive (pH, thermal, magnetic) composites, Nanocomposites for targeted drug delivery; Biodegradable composite microspheres and scaffolds, Smart composites for health monitoring; Theranostic composite systems.

Characterisation and Testing of Biomedical Composites (7 lectures): Mechanical testing standards for biomedical composites, Tribological testing, Surface characterisation, clean room, animal testing, release, migration testing.

Textbooks

1) Biomedical Composites, L. Ambrosio (Ed.), Woodhead Publishing, 2nd Ed. (2013).
2) Biomaterials Science: An Introduction to Materials in Medicine, Ratner, Hoffman, Schoen, Lemons, Elsevier (2013).

Reference Books

1) An Introduction to Biocomposites, S. Ramakrishna et al., Imperial College Press (2004).
2) Composite Materials for Biomedical Applications, M. Ramalingam et al. (Eds.), ASM International (2012).
3) ASTM F2026, F2083, ISO 10993 series (Biocompatibility evaluation standards).

Course Title	Composites for Energy Harvesting and Storage	Course No.	MTL7XXX
Department	Materials Engineering	Structure (L-T-P-D)	3-0-0

Offered for	M.Tech./Ph.D.	Type	Elective
Prerequisite	-		

Learning Objectives:

The objectives of this course are to:

1. Develop understanding of the role of composite materials in advancing energy harvesting and storage technologies.
2. Enable students to analyse structure–property relationships in electrode, electrolyte, and transducer composite systems for sustainable energy applications.

Learning Outcomes:

After completing this course, students will be able to:

1. Correlate composition, microstructure, and architecture of composites with their electrochemical and piezoelectric performance.
2. Design and evaluate composite systems for batteries, supercapacitors, fuel cells, solar cells, and piezoelectric energy harvesters.

Contents

Fundamentals of Energy Materials and Composites (5 lectures): Energy landscape and sustainability drivers; Classification of energy composites; Electrochemical fundamentals: charge transport, diffusion, impedance; Piezoelectric, thermoelectric, and triboelectric mechanisms; multi-scale design of energy composites; Characterization overview.

Composites for Lithium-Ion and Next-Generation Batteries (8 lectures): Electrode composite architectures: anode and cathode systems; Binder and conductive additive role; Solid electrolyte composites, Composite separators; All-solid-state battery composites; Sodium-ion and lithium–sulfur composite electrode design; Capacity fade and failure mechanisms.

Composites for Supercapacitors (6 lectures): Electrochemical double-layer capacitors and pseudocapacitors, Carbon–metal oxide composites, Conductive polymer composite electrodes; Hybrid supercapacitor design; Gel polymer composite electrolytes; Rate capability, cycling stability, and scalability.

Composites for Fuel Cells and Solar Energy (7 lectures): Polymer electrolyte membrane composites: Nafion composite membranes, catalyst layer composites; Solid oxide fuel cell (SOFC) electrode composites; Photovoltaic composites: bulk heterojunction polymer–fullerene, perovskite composites; Dye-sensitised solar cell, photoanode composites; Anti-reflection and encapsulant composites.

Piezoelectric, Thermoelectric, and Triboelectric Composites (7 lectures): Piezoelectric polymer–ceramic composites, Connectivity patterns and coupling coefficients; Thermoelectric composite thin films and bulk nanocomposites; Figure of merit enhancement strategies; Triboelectric nanogenerator composite surfaces; Flexible and wearable energy harvesting composites.

Testing, Reliability, and Emerging Trends (6 lectures): Electrochemical characterisation: cyclic voltammetry, galvanostatic cycling, EIS; Mechanical–electrochemical coupling and swelling; Thermal management composites for energy devices; Self-healing composite electrodes; Multifunctional structural energy storage composites; Life cycle assessment and sustainability.

Textbooks

1) Nanocomposites for Electrochemical Energy Storage, A. Banerjee (Ed.), Wiley (2020).
2) Piezoelectric Transducers and Applications, A.A. Vives (Ed.), Springer (2008).

Reference Books

1) Carbon Nanomaterials for Electrochemical Energy Conversion and Storage, J. Liu et al., Wiley-VCH (2019).
2) Fuel Cell Fundamentals, O'Hayre, Cha, Colella, Prinz, Wiley (2016).
3) Selected review articles from Journal of Power Sources, Advanced Energy Materials, and Nano Energy.

Course Title	Machining and Tribology of Composites	Course No.	MTL7XXX
Department	Materials Engineering	Structure (L-T-P-D)	3-0-0

Offered for	M.Tech./Ph.D.	Type	Elective
Prerequisite	-		

Learning Objectives:

The objectives of this course are to:

1. Equip students with knowledge of machining science and tribological behaviour unique to composite materials.
2. Develop ability to select optimal machining processes, cutting tools, and lubricating strategies to minimise damage and surface degradation in composite components.

Learning Outcomes:

After completing this course, students will be able to:

1. Analyse chip formation, cutting forces, and surface integrity during machining of polymer, metal, and ceramic matrix composites.
2. Evaluate wear mechanisms, friction behaviour, and tribological performance of composites in dry, lubricated, and abrasive contact conditions

Contents

Introduction to Machining of Composite Materials (5 lectures): Challenges of machining heterogeneous anisotropic materials; Fibre orientation effects on chip formation; Tool–workpiece interaction; Overview of machining processes, Machinability index; Surface integrity

Machining of Polymer Matrix Composites (8 lectures): Drilling of CFRP and GFRP: thrust force, delamination, peel-up/push-down mechanisms; Delamination factor and damage assessment; Tool wear in CFRP, Trimming and edge routing; Milling of thermoplastic composites; Abrasive water jet machining,Ultrasonic-assisted machining

Machining of Metal and Ceramic Matrix Composites (8 lectures): Machining of Al-SiC and Ti-based MMCs: chip segmentation, built-up edge; CBN and PCD tool performance in MMC machining; Machining of CMCs: SiC/SiC and C/C composites; Grinding as primary shaping process for CMCs; Laser-assisted machining, Surface damage and strength degradation after machining.

Tribology of Polymer Matrix Composites (6 lectures): Friction and wear mechanisms in PMCs; Role of fibre type, orientation, and volume fraction; Solid lubricant filler, Dry sliding, rolling, and erosive wear; Tribological behaviour of short fibre, hybrid, and nanoparticle-filled composites; PV limit and temperature rise modelling.

Tribology of Metal and Ceramic Matrix Composites (6 lectures): Adhesive, abrasive, and fatigue wear in MMC brake and bearing systems; SiC particulate reinforcement effects on wear rate; Tribology of C/C composites for brake applications, Oxidation–wear coupling; Ceramic coating composites for wear protection; High-temperature tribology

Surface Engineering and Quality Assessment (6 lectures): Coatings and surface treatments for composite tribological improvement, Profilometry and roughness measurement; Acoustic emission and vibration monitoring during machining; Chip morphology analysis; NDT, Sustainability, dry machining.

Textbooks

1) Machining of Fibre-Reinforced Plastics, D. Hocheng (Ed.), Springer (2012).
2) Tribology of Polymeric Nanocomposites, K. Friedrich & A.K. Schlarb (Eds.), Elsevier (2013).

Reference Books

1) Machining of Metal Matrix Composites, J.P. Davim (Ed.), Springer (2012).
2) Principles of Tribology, J. Williams, Wiley (2005).
3) Selected articles from Wear, Tribology International, and Composite Structures.

Course Title	Advanced Composite Design and Analysis	Course No.	MTL7XXX
Department	Materials Engineering	Structure (L-T-P-D)	3-0-0
Offered for	M.Tech./Ph.D.	Type	Elective
Prerequisite	-		

Learning Objectives:

The objectives of this course are to:

1. Develop advanced analytical and computational skills for designing composite structural components.
2. Enable students to apply laminate theory, failure criteria, and numerical methods to optimise composite structures for aerospace, automotive, and civil applications.

Learning Outcomes:

After completing this course, students will be able to:

1. Apply classical laminate plate theory and first-ply failure criteria to multi-directional laminates.
2. Use finite element analysis and optimisation techniques for weight-efficient composite structural design, and evaluate joints, connections, and repairability of composite assemblies

Contents

Advanced design for composite structures (8 Lectures)

Applications of advanced composites in structures, cost of composites, technology Selection, composite structural components, Structural Design Process and Analyst's Perspective, Design Concept and Material/Process Considerations, Uncertainty in Composite Design, Environmental Effects and Damage Considerations, Design values and allowables, Good design practices and Rules of Thumb

Micromechanics and Constitutive Relations (6 lectures): Anisotropic elasticity: compliance and stiffness tensors; Micromechanical models (Voigt, Reuss, Halpin–Tsai, Mori–Tanaka); Effective moduli prediction and bounds; Hygrothermal constitutive relations; Viscoelastic behaviour of composites; Experimental validation of micromechanical predictions.

Classical Laminate Plate Theory (CLPT) (7 lectures): Kinematics of thin plates; Plane stress assumption; Transformed reduced stiffness; [A], [B], [D] matrices; Symmetric, antisymmetric, and balanced laminates; Hygrothermal loads and residual stresses; Free-edge effects and interlaminar stresses; First-ply failure and last-ply failure sequences

Failure Criteria and Damage Mechanics (6 lectures): Maximum stress and maximum strain criteria, Tsai–Hill and Tsai–Wu polynomial criteria, Continuum damage mechanics (CDM) framework; Fatigue damage evolution in laminates, Impact damage tolerance assessment.

Stability, Dynamics, and Optimization of Composite Structures (7 lectures): Buckling of composite plates and columns: Navier and Rayleigh–Ritz solutions; Post-buckling and imperfection sensitivity; Vibration analysis: natural frequencies and mode shapes; Aeroelastic considerations; Topology and fibre angle optimisation, Genetic algorithm and gradient-based methods; Multi-objective and multi-constraint design.

Joints, Repairs, and Structural Health Monitoring (5 lectures): Mechanical and adhesive joint design for composites, Repair qualification criteria, Embedded fibre Bragg grating sensor, Digital twin concepts for composite structures.

Textbooks

1) Mechanics of Composite Materials, R.M. Jones, Taylor & Francis (1999).
2) Principles of Composite Material Mechanics, R.F. Gibson, CRC Press, 4th Ed. (2016

Reference Books

1) Analysis and Performance of Fibre Composites, B.D. Agarwal, L.J. Broutman & K. Chandrashekhara, Wiley (2006).
2) Composite Structures: Safety Management, B. Tatting & Z. Gürdal, Elsevier (2004).
3) ANSYS Mechanical APDL Composite Analysis Guide; Abaqus Analysis User's Manual – Composites.

Course Title	Composites for Defence Technology	Course No.	MTL7XXX
Department	Materials Engineering	Structure (L-T-P-D)	3-0-0
Offered for	M.Tech./Ph.D.	Type	Elective
Prerequisite	-		

Learning Objectives:

The objectives of this course are to:

2. Provide comprehensive knowledge of composite materials engineered for defence, armour, and stealth applications.
3. Develop students' ability to evaluate multi-threat protection requirements and apply composite science to ballistic, blast, electromagnetic, and signature-management challenges.

Learning Outcomes:

After completing this course, students will be able to:

1. Analyse ballistic and blast response mechanisms of structural composites and design multi-layered protective systems.
2. Evaluate radar-absorbing, EMI-shielding, and infrared-signature-reduction composites for stealth platforms.

Contents

Introduction to Defence Composites (5 lectures): Role of advanced composites in modern defence systems, Weight reduction and performance drivers, Overview of defence platforms, Classification of structural, functional, and multifunctional defence composites, Export control and classified material regulations.

Ballistic Protection Composites (8 lectures): Projectile–target interaction mechanics, Energy absorption mechanisms, High-performance fibres, confinement effects, dwell time; Transparent armour composite, ballistic protection standards, Residual velocity and V50 testing.

Blast and Shock Loading of Composites (7 lectures): Blast wave characterisation: Friedlander waveform, overpressure; Sandwich composite panels, Fibre metal laminates for blast mitigation, Underwater shock response of marine composites, Energy-absorbing composite structures, Mine-resistant underbelly composites.

Stealth and Electromagnetic Composites (8 lectures): Radar cross-section reduction principles, Infrared signature management, low-emissivity composite coatings, EMI shielding composites, Shielding effectiveness measurement, Acoustic stealth composites.

Structural Composites for Defence Platforms (6 lectures): Aerospace defence composites. fighter aircraft fuselage, wing, and rotor blade, Naval composite superstructures and masts, High-temperature composites for missile airframes: C/C and CMC applications, Composite torpedo and unmanned vehicle structures, environmental durability in service.

Testing, Qualification, and Future Directions (5 lectures): Military testing standards: MIL-SPEC, DEF STAN, Non-destructive inspection of defence composites, Accelerated ageing and environmental qualification; Multi-threat testing protocols; Self-healing composites for damage tolerance, Multi-functional composites, Advances in composites

Textbooks

1. Lightweight Ballistic Composites: Military and Law Enforcement Applications, A. Bhatnagar (Ed.), Woodhead Publishing, 2nd Ed. (2016).
2. Blast Protection of Civil Infrastructures and Vehicles Using Composites, N. Uddin (Ed.), Woodhead Publishing (2010).

Reference Books

1) Armour: Materials, Theory, and Design, P.C. Chou, Technomic (1992).
2) Composite Materials for Aerospace Applications, B.D. Agarwal, CRC Press (2019).
3) Selected articles from International Journal of Impact Engineering, Composite Structures, and Defence Technology.

Course Title	Functional Nanocomposites and Hybrid Composites	Course No.	MTL7XXX
Department	Materials Engineering	Structure (L-T-P-D)	3-0-0
Offered for	M.Tech./Ph.D.	Type	Elective
Prerequisite	-		

Learning Objectives:

The objectives of this course are to:

1. Develop a thorough understanding of nano-scale reinforcement mechanisms and the design of hybrid composite systems.
2. Enable students to engineer nanocomposites and hybrid composites with tailored electrical, thermal, magnetic, optical, and mechanical functionalities

Learning Outcomes:

After completing this course, students will be able to:

1. Explain how nano-scale architecture governs properties of nanocomposites and predict property enhancement using mixing rules and percolation theory.
2. Design and characterise multifunctional nanocomposite and hybrid composite systems for sensing, actuation, thermal management, and electromagnetic applications.

Contents

Fundamentals of Nanoscale Reinforcement (6 lectures): Size-dependent properties at nano-scale, Nano-reinforcement types, Surface area, aspect ratio, and interfacial zone concepts, Filler–matrix interface chemistry and functionalisation.
Carbon-Based Nanocomposites (6 lectures): Carbon nanotubes, polymer nanocomposites, Graphene and graphene oxide (GO, rGO) nanocomposites, Multiscale composites, Buckypaper composite films, Challenges.
Inorganic Nanoparticle and Clay Nanocomposites (8 lectures): Intercalation and exfoliation, Polymer–clay nanocomposites, Metal oxide nanoparticles, Nanosilica and nanoalumina in rubber and polymer matrices, POSS composites, Quantum dot nanocomposites.
Hybrid Composite Systems (7 lectures): Definition and classification of hybrid composite, Fibre hybridisation, Hybrid rule of mixtures and failure envelope, Hybrid sandwich and functionally graded composites, Bio-based hybrid composites.
Multifunctional and Smart Nanocomposites (6 lectures): Electrically conductive nanocomposites, Shape memory polymer nanocomposites, Self-healing nanocomposites, Piezoelectric nanocomposite, Photocatalytic and antimicrobial nanocomposite surfaces
Processing, Characterisation, and Applications (5 lectures): Dispersion techniques, Processing challenges and quality control in nanocomposites, Characterization, Nanocomposite coatings and thin films, Safety, toxicology, and regulatory aspects of nanomaterials.

Textbooks

1) Polymer Nanocomposites, Y.-W. Mai & Z.-Z. Yu (Eds.), Woodhead Publishing (2006).
2) Hybrid Composites, J.P. Davim & M. Charitidis (Eds.), De Gruyter (2021).

Reference Books

1) Carbon Nanotube–Polymer Composites, B. Bhattacharya, Wiley (2012).
2) Clay-Containing Polymer Nanocomposites, S. Sinha Ray, Elsevier (2013).
3) Selected articles from Composites Science and Technology, ACS Nano, and Small.